

AIDS Project Proposal

India's EV Transition State-wise Adoption Trends & Forecasting
Group 11

1. Topic & Significance

This project analyses India's transition to electric vehicles (EVs) using monthly fuel-type wise vehicle registration data and state-wise EV registration data, both sourced from data.gov.in. India has set ambitious targets for EV adoption 30% of new vehicle sales to be electric by 2030 under the FAME scheme yet adoption remains highly uneven across states and vehicle categories. Understanding where adoption is accelerating, which states are leading, and how the fuel mix is shifting month by month is critical for policy-making, infrastructure planning, and industry forecasting.

A key metric in this transition is the EV share the percentage of total vehicle registrations that are electric. We aim to study how this share varies across states and time, model the growth trend, and forecast when EVs will cross 10% of total national registrations.

2. Dataset

Source: data.gov.in Ministry of Road Transport & Highways (MoRTH), Government of India.

Dataset 1 Fuel-type wise Vehicle Registrations (All India Monthly, 2014–2024): Monthly all-India vehicle registrations broken down by vehicle class and fuel type (Petrol, Diesel, CNG, Electric, Hybrid, etc.). This allows us to track the national fuel mix shift over a decade.

Link:

data.gov.in/resource/all-india-level-vehicle-class-and-fuel-type-wise-total-number-vehicles-registered-each

Dataset 2 State/UT-wise EV Registrations % (2019–2024): State and UT-wise total number of EVs registered and their percentage share of total vehicle registrations. This allows comparison of EV adoption across all Indian states.

Link:

data.gov.in/resource/stateut-wise-total-number-electric-vehicles-evs-registered-percentage-total-number

Key Columns: State/UT, Year, Month, Fuel Type, Vehicle Class, Total Registrations, EV Registrations, EV Share (%).

3. Random Variables

X1 Monthly EV Registrations (National): Total number of electric vehicles registered per month across India. This is the primary measure of EV demand and captures the growth trajectory over 2014–2024.

X2 EV Share (%): Percentage of total vehicle registrations that are electric, computed per state per year. This is the key variable for cross-state comparison and the target variable for our 10% threshold forecast.

X3 Fuel Mix Ratio: Ratio of EV + CNG + Hybrid registrations to total registrations per month. This is a broader measure of the clean fuel transition beyond just EVs.

4. Proposed Hypotheses

H1 Leadership Gap: The mean annual EV share of the top 3 EV-adopting states (e.g., Goa, Delhi, Kerala) is significantly higher than the national mean EV share over 2019–2024.

H2 Post-FAME Acceleration: Mean monthly EV registrations in the post-FAME II period (2019–2024) are significantly higher than in the pre-FAME II period (2014–2018), indicating that the policy had a measurable impact on adoption.

H3 Consistency of Growth: States with higher per-capita income have significantly lower variance in year-on-year EV share growth, indicating more stable and predictable adoption patterns compared to lower-income states.

5. Tentative Analysis Steps

1. **Data Cleaning and Preparation:** Filter to relevant fuel types, handle missing monthly entries, compute EV share per state per year, and create derived variables such as pre/post-FAME II period flags and EV growth rate year-on-year.
2. **Exploratory Visualisation:** Plot national monthly EV registration time-series (2014–2024), bar charts of state-wise EV share, a ranked comparison of states by adoption rate, and a fuel mix stacked area chart showing the shift from Petrol/Diesel toward cleaner fuels over time.
3. **Distribution Fitting:** Fit candidate distributions (Log-Normal, Exponential, Gaussian) to X1 and X2. Assess goodness-of-fit using Q-Q plots and the Kolmogorov–Smirnov test.
4. **Hypothesis Testing:** Formally test H1, H2, and H3. Report the test statistic, p -value, and conclusion at the 5% significance level for each hypothesis. Interpret the results in the context of India’s EV transition.
5. **Correlation and Regression Analysis:** Compute correlation between time (month index) and national EV share. Perform simple linear regression of EV share on time. Extend to an exponential or logistic growth model if the linear fit is poor.
6. **Forecasting the 10% Threshold:** Using the fitted regression or growth model, forecast when national EV share will cross 10% of total registrations.
7. **Summary and Conclusions:** Summarise which states are leading, whether FAME II had a statistically significant effect, and the forecasted timeline for the 10% EV threshold at the national level.

6. Work Distribution

Member	Tasks	Roll No.
Ravindra Bandaru	Data cleaning, exploratory visualisation, fuel mix time-series, distribution fitting, Q-Q plots	24B0006
Malla Jaswanth Narayana	Hypothesis formulation and testing, significance analysis, FAME II impact assessment, video narration	24B0022
Songa Sai Balaji	Correlation analysis, regression, growth model, 10% threshold forecasting, conclusions, documentation	24B0047
Chinthakunta Mahith Kumar Reddy	Data preprocessing support, additional visualisations, validation checks, documentation review and presentation preparation	24B0030